

Yuan-Hung Kuan

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Education

Washington University in St. Louis | Ph.D.

St. Louis, MO

- *Ph.D. in Electrical and Systems Engineering (ESE)*
- *Systems Science and Mathematics Program* | Advisor: Professor Jr-Shin Li
- GPA: 3.82/4.0 | McDonnell International Scholars Academy

Sep 2020 - present

National Taiwan University | B.S.

Taipei, Taiwan

- *Major in Bioenvironmental Systems Engineering (BSE)*
- GPA: 3.39/4.0 | *Minor in Mathematics (MATH)*

Sep 2015 - Jun 2019

Presentations / Talks

Invited Talk

- Seminar talk, Los Alamos National Laboratory (Los Alamos, Feb 10, 2026): “*Structure and Scalable Representations for Ensemble Quantum and Dynamical Systems.*”
- Invited workshop presentation, IEEE CDC 2023 (Singapore, Dec 12-15, 2023): “*Learning for Dynamical Systems and Ensemble Systems using Reservoir Computing Networks.*”
- Panel discussion on mixed research topics (Taipei, Jun, 2024).

Oral Presentation

- Regular session presentation, SIAM CT25 (Montreal, Jul 28-30, 2025): presenting [J2].
- Regular session presentation, IEEE CDC 2024 (Milan, Dec 16-19, 2024): presenting [C1].
- Regular session presentation, IEEE CDC 2023 (Singapore, Dec 12-15, 2023): presenting [C2].
- Seminar talk, System Science Seminar at Washington University in St. Louis (Oct 2, 2025): presenting [P1] “*EBIF: Exact Bilinearization Iterative Form for Control-Affine Nonlinear Systems.*”
- Undergraduate summer intern seminar talk at Washington University in St. Louis (Aug 28, 2019): “*Inferring Connections Of Networks & manifold learning using semidefinite programming.*”

Poster Presentation

- The Lusted Student Award Competition (Finalist), SMDM 2024 (Boston, Oct 27-30, 2024): presenting [M1].
- Poster presentation, SMDM 2024 (Boston, Oct 27-30, 2024): presenting [M4].
- Poster presentation, I2DB 2024 (St. Louis, Apr 12, 2024): “*Using ML to identify dynamic markers for multiple myeloma in patients with diabetes mellitus and monoclonal gammopathy of undetermined significance.*”
- Poster presentation, ICTS 2024 (St. Louis, May 17, 2024): “*Prediction of in-hospital mortality using artificial intelligence to handle incomplete, and time-varying Intensive Care Unit data.*”
- Poster presentation at Washington University in St. Louis (Feb 4, 2023): “*An Iterative Reservoir Computing Method to Reconstruct Irregular and Sparse EHR Data.*”

SIAM CT25: SIAM Conference on Control and Its Applications; IEEE CDC: IEEE Conference on Decision and Control conference; SMDM: The Society for Medical Decision Making Annual North American Meeting; I2DB: Institute for Informatics, Data Science & Biostatistics 2024; ICTS: The Washington University Institute of Clinical and Translational Sciences’ annual symposium

Honors & Awards

Lee B. Lusted Student Award Finalist

2024

- in *The 46th Society for Medical Decision Making Annual North American Meeting (SMDM)*

The McKelvey Engineering Professional Development Award

2024

The Outstanding Graduate Student Assistant to the Instructor Award

2023

The McKelvey Engineering Professional Development Award

2023

McDonnell International Scholars Academy

2020

Dean’s Select PhD Scholar

2020

Taiwan Ministry of Education Fellow

2020

Citation Summary

Citation counts for individual publications are listed below. Counts include archived manuscripts/preprints when indexed by Google Scholar.

◦ Citation source: **Google Scholar** retrieved on *March 19, 2026* | **Total citations: 24**

Publications

Journal:

- (J1.) **Y.-H. Kuan**, V. Narayanan and J.-S. Li, “Iterative Reservoir Computing Networks for Reconstructing Irregular Time Series,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 36, no. 8, pp. 14189-14200, Aug. 2025, doi: [10.1109/TNNLS.2025.3547965](https://doi.org/10.1109/TNNLS.2025.3547965) . [Google Scholar Citations: 1]
- (J2.) J.-S. Li, **Y.-H. Kuan**, W. Zhang, “Optimal Quantum Control using Ensemble Quantization,” *SIAM Journal on Control and Optimization*, vol. 63, no. 1, pp. S107-S127, Feb. 2025, doi: [10.1137/23M1590809](https://doi.org/10.1137/23M1590809) . [Google Scholar Citations: 3]
- (J3.) J.-S. Li, W. Zhang, **Y.-H. Kuan**, “Moment quantization of inhomogeneous spin ensembles,” *Annual Review in Controls*, vol. 54, pp. 305-313, ISSN 1367-5788, Aug. 2022, doi: [10.1016/j.arcontrol.2022.07.006](https://doi.org/10.1016/j.arcontrol.2022.07.006) . [Google Scholar Citations: 14]
- (J4.) M. Wang, **Y.-H. Kuan**, J.-S. Li, S.-H. Chang, “Identification of dynamic markers for multiple myeloma in patients with MGUS and T2DM,” *Blood*, vol. 146, suppl. 1, pp. 5784, ISSN 0006-4971, Dec. 2025, doi: [10.1182/blood-2025-5784](https://doi.org/10.1182/blood-2025-5784) .

Conference:

- (C1.) **Y.-H. Kuan**, W. Zhang, J.-S. Li, “Computational Moment Control of Ensemble Systems,” *2024 63rd IEEE Conference on Decision and Control*, Milan, Italy, 2024, pp. 1251-1256, 2024, doi: [10.1109/CDC56724.2024.10886536](https://doi.org/10.1109/CDC56724.2024.10886536) . [Google Scholar Citations: 3]
- (C2.) **Y.-H. Kuan**, X. Ning, J.-S. Li, “A Generalized-Moment Method for Control-Affine Ensemble Systems,” *2023 62nd IEEE Conference on Decision and Control*, Singapore, Singapore, 2023, pp. 1654-1659, doi: [10.1109/CDC49753.2023.10383974](https://doi.org/10.1109/CDC49753.2023.10383974) . [Google Scholar Citations: 3]
- (C3.) M. Wang, **Y.-H. Kuan**, P. Alba, Q. Gan, M.W. Schoen, T.S. Thomas, J.-S. Li, S.-H. Chang, “Developing Large Language Model-based Pipeline for Identification of Disease Diagnosis: A Case Study on Identifying Newly Diagnosed Multiple Myeloma in Veterans Health Administration Electronic Health Records,” *AMIA Annual Symposium Proceedings*, vol. 2024, pp. 1325-1334, PMID: [41726535](https://pubmed.ncbi.nlm.nih.gov/41726535/)  ; PMCID: [PMC12919547](https://pubmed.ncbi.nlm.nih.gov/PMC12919547/) , May 2025.
- (C4.) W. Zhang, **Y.-H. Kuan**, S.-H. Chang, J.-S. Li, “A Deep Reservoir Computing Architecture for Dynamic Generative Modeling,” *2025 International Joint Conference on Neural Networks (IJCNN)*, Rome, Italy, 2025, pp. 1-8, doi: [10.1109/IJCNN64981.2025.11227706](https://doi.org/10.1109/IJCNN64981.2025.11227706) .

Meeting:

- (M1.) **Y.-H. Kuan**, M. Wang, J.-S. Li, S.-H. Chang, “A novel deep reservoir computing method to impute electrocardiography for risk prediction models assisting medical decision making,” *46th Annual Meeting of the Society for Medical Decision Making*, Boston, Massachusetts, United States, 26-30 October 2024. Medical Decision Making. 2025;45(5):NP1-NP330, doi: [10.1177/0272989X251347263](https://doi.org/10.1177/0272989X251347263)  (Poster Presentation; Lee B. Lusted Student Award Finalist).
- (M2.) M. Wang, **Y.-H. Kuan**, J.-S. Li, S.-H. Chang, “Leveraging natural language processing to identify smoldering multiple myeloma from electronic health record data,” *47th Annual Meeting of the Society for Medical Decision Making*, Ann Arbor, Michigan, United States, 15-18 June 2025. Medical Decision Making. 2025;45(7):NP1-NP244, doi: [10.1177/0272989X251361142](https://doi.org/10.1177/0272989X251361142)  (Poster Presentation).
- (M3.) M. Wang, **Y.-H. Kuan**, J.-S. Li, S.-H. Chang, “Identifying dynamic markers for multiple myeloma in patients with diabetes mellitus and monoclonal gammopathy of undetermined significance,” *46th Annual Meeting of the Society for Medical Decision Making*, Boston, Massachusetts, United States, 26-30 October 2024. Medical Decision Making. 2025;45(5):NP1-NP330, doi: [10.1177/0272989X251347263](https://doi.org/10.1177/0272989X251347263)  (Oral Presentation).

- (M4.) **Y.-H. Kuan**, M. Wang, J.-S. Li, S.-H. Chang, “Using deep neural networks to process incomplete time-varying Intensive Care Unit data for in-hospital mortality prediction,” *46th Annual Meeting of the Society for Medical Decision Making*, Boston, Massachusetts, United States, 26-30 October 2024. Medical Decision Making. 2025;45(5):NP1-NP330, doi: [10.1177/0272989X251347263](https://doi.org/10.1177/0272989X251347263)  (Poster Presentation).
- (M5.) **Y.-H. Kuan**, J.-S. Li, S.-H. Chang, “An iterative reservoir computing method to reconstruct irregular and sparse EHR data,” *45th Annual North American Meeting of the Society for Medical Decision Making*, Philadelphia, Pennsylvania, United States, 22-25 October 2023. Medical Decision Making. 2024;44(3):NP1-NP375, doi: [10.1177/0272989X241236400](https://doi.org/10.1177/0272989X241236400)  (Poster Presentation).

Submitted Paper:

- (S1.) **Y.-H. Kuan**, X. Liang, and J.-S. Li, “Zonotope-based Reachability Analysis for Nonlinear Systems through Bilinear Embedding,” under review at *the 65th IEEE Conference on Decision and Control (CDC, cdc2026.ieeecss.org)* .
- (S2.) **Y.-H. Kuan**, L. Tang, and J.-S. Li, “Reservoir Computing Causal Operators for Data-Driven Moment Control of Nonlinear Ensembles,” under review at *the IEEE Control Systems Letters (L-CSS, ieee-control-systems-letters)* .
- (S3.) **Y.-H. Kuan**, and J.-S. Li, “Quantum Ensemble Control in Bose–Einstein Condensates,” under review at *the 2026 IEEE International Conference on Quantum Control, Computing, and Learning (IEEE qCCL 2026, www.ieeeqcl2026.com)* .
- (S4.) **Y.-H. Kuan**, J.-S. Li, “Geometric Optimal Control of Nonlinear Systems through Bilinear Embedding,” under review at *the International Federation of Automatic Control (IFAC) The 23rd International Federation of Automatic Control World Congress (IFAC WC 2026, ifac2026.org/fairDash.do?hl=ENG)* .
- (S5.) J.-S. Li, M. Yang, **Y.-H. Kuan**, and W. Zhang, “Path Control of Linear Ensemble Systems on Hilbert Spaces,” under review at *the International Federation of Automatic Control (IFAC) The 23rd International Federation of Automatic Control World Congress (IFAC WC 2026, ifac2026.org/fairDash.do?hl=ENG)* .
- (S6.) H. Quan, W. Zhang, **Y.-H. Kuan**, C. Sun, J.-S. Li, “Pattern Control in Robot Swarms through Dynamic Reduction,” under review at *the International Journal of Robotics Research (IJRR, journals.sagepub.com/home/IJRR)* .

Preprints / Ongoing Manuscripts:

- (P1.) **Y.-H. Kuan**, J.-S. Li, “EBIF: Exact Bilinearization Iterative Form for Control-Affine Nonlinear Systems,” manuscript in preparation, expected submission to *IEEE Transactions on Automatic Control (TAC, transactions-automatic-control)* ; preprint available at *arXiv:2512.17036*, Dec. 2025, doi: [10.48550/arXiv.2512.17036](https://doi.org/10.48550/arXiv.2512.17036) .
- (P2.) **Y.-H. Kuan**, W. Zhang, V. Narayanan, W.-Y. Shen, H. Xiao, J.-S. Li, “Layer-Varying Deep Reservoir Computing Architecture,” *OpenReview*, 2025, doi: openreview.net/forum?id=qVtfN6NoJi .
- (P3.) **Y.-H. Kuan**, J. Fischbach, J.-S. Li, K. Krischer, “Optimal Control Problem on Dynamic Catalysis,” manuscript in preparation, expected submission to *SIAM Journal on Applied Dynamical Systems (SIADS, siam-journal-on-applied-dynamical-systems/)* .

Teaching Experience

Graduate Teaching Assistant (TA)

2021-2026

- ESE 415 *Optimization* (Fall 2021, Fall 2022, Spring 2026)
- ESE 551 *Linear Dynamic Systems* (Fall 2024, Fall 2025)
- ESE 5581 *Advanced Systems Science for Learning and Control of Complex Dynamic Systems* (Spring 2023, Spring 2025)

Selected responsibilities:

- Delivered lectures on: *review lectures on linear algebra & linear operator, optimization methods, optimal control, reservoir computing, review lectures on optimization & linear control systems*
- Supported course instruction, problem sessions, and student mentoring

Research Experience

Graduate Research Assistant

Washington University in St. Louis

St. Louis, MO

Sep 2020 - present

- Conduct research in control theory, quantum systems and control, ensemble systems, bilinear representations, reservoir computing, and learning for dynamical systems.
- Developed methods for exact bilinearization, moment-based ensemble control, iterative and deep reservoir architecture, and reachability analysis of nonlinear systems.
- Collaborated on interdisciplinary projects in medical AI and clinical natural language processing using electronic health record data.

Undergraduate Summer Research Internship

Washington University in St. Louis

St. Louis, MO

Jul 2019 - Sep 2019

- Completed an undergraduate summer research internship in the department of Electrical and Systems Engineering under the guidance of professor Jr-Shin Li.
- Title: *Inferring Connections Of Networks & manifold learning using semidefinite programming*

Research Assistant

National Taiwan University

Taipei, Taiwan

Apr 2020 - Jun 2020

- Conduct research in simplex projection methods for time series analysis

Professional Services

Professional societies: Institute of Electrical and Electronics Engineers (IEEE) student member; Society for Industrial and Applied Mathematics (SIAM) student member

Manuscript Reviewer for: Scientific Reports; International Conference on Learning Representations (ICLR); IEEE Conference on Decision and Control (CDC); International Federation of Automatic Control (IFAC); IEEE International Conference on Quantum Control, Computing and Learning (QCCL)

Technical Skills

Research areas: Control theory, Ensemble systems and control, Quantum control, Numerical analysis & optimization, Machine learning, Deep learning, Large language models, Visualization

Programming Languages: MATLAB, Python, LaTeX, R

Frameworks: PyTorch, Matplotlib, Keras, Pandas, Scikit-learn, NumPy, SciPy, GitHub

Languages: English (Fluent) and Chinese (Native)

Other Experience (Relevant Extracurricular Activities)

I2DB Datathon Competition

Washington University in St. Louis

St. Louis, MO

Mar 2024 - Apr 2024

PhD-in-3D Competition

Washington University in St. Louis

St. Louis, MO

Mar 2024

Global Leadership Visions Presentation

Washington University in St. Louis

St. Louis, MO

Feb 23, 2022

Undergraduate Summer Research Internship Student Leader

Washington University in St. Louis

St. Louis, MO

Jun 2024 - Aug 2024

Undergraduate Summer Research Internship Student Leader

Washington University in St. Louis

St. Louis, MO

Jun 2023 - Aug 2023